

CASE STUDY: REDUCING MACHINING DEFECTS

Stabilizing CNC Operations through Standardized Setup Controls

Executive Summary

Dimensional inaccuracies during CNC machining (FOM) were causing significant over-milling and rework due to extrusion bowing. By transitioning from subjective operator adjustments to a structured **"First-Off" Inspection** and periodic verification system, I stabilized the process, reduced dimensional variation, and eliminated reliance on individual judgment.

1. CONTEXT & CHALLENGE

The machining of long extrusions was plagued by bowing, which led to inconsistent clamping and over-milling. These dimensional defects were often discovered too late in the process, resulting in high scrap rates and assembly delays.

- **The Problem:** Process stability was overly dependent on individual operator skill rather than a standardized system.
- **Technical Barriers:** * Inconsistent clamping conditions across different shifts.
 - Absence of a repeatable method to measure extrusion bow.
 - Lack of a mid-process verification cadence.

2. STRATEGIC APPROACH (SYSTEMS THINKING)

I approached the issue as a **Process Stability Failure** rather than an operator error.

- **Gemba Observation:** Spent time at the FOM machine to identify how setup variation and clamping pressure contributed to material deflection.
- **Variation Analysis:** Mapped the points where manual "corrections" were actually introducing more instability into the system.
- **Root Cause:** Identified that the lack of a "Control Point" at the start of the shift allowed defects to cascade through the entire batch.

3. THE SOLUTION: REPEATABLE CONTROL STRUCTURE

I introduced a multi-layered control framework to lock in dimensional accuracy:

1. **First-Off Inspection System:** Mandatory dimensional confirmation at the start of every run to validate setup integrity.
2. **Periodic QC Cadence:** Implemented a 2-hour verification window to catch "process drift" before it resulted in scrap.
3. **Standardized Clamping & Offsets:** Optimized the offset strategy and proposed a custom fixture to ensure consistent bow measurement.
4. **Stock Optimization:** Provided recommendations for material handling to minimize bowing before the extrusion even reached the machine.

4. QUANTIFIED IMPACT

Key Performance Indicator	Baseline (Pre-Intervention)	Post-Intervention Result
Dimensional Consistency	Variable / Unstable	High / Repeatable
Machining Defects	High Volume	Significantly Reduced
Process Control	Operator-Dependent	System-Governed
Rework & Scrap	Frequent / Costly	Minimized

5. LEADERSHIP TAKEAWAYS

- **Control at Source:** High-precision manufacturing requires variation to be managed at the setup stage, not corrected after the fact.
- **Standardization Over Judgment:** System reliability improves when you replace "individual intuition" with "measurable verification."
- **Cadence Matters:** Short feedback loops (2-hour checks) are the most effective way to stabilize high-variation operations.

Prepared by: Hussain Syed | P.Eng, PMP

Expertise: CNC Process Optimization & Quality Governance